Hoz Serkany

Computer Engineering

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Computer engineering professional with production experience building operator-facing command-and-control interfaces, live telemetry dashboards, geospatial visualizations, and sensor-integrated web platforms. Delivered Node.js BFF services, Redux/Next.js frontends, and AWS (ECS, EKS, Lambda, S3) deployments at Sensofusion; embedded Linux and CI/CD automation at Ericsson. Founder of Şandin Tech Inc., leading full-stack product engineering from architecture through launch. Experienced in field deployments and cross-functional hardware/software integration.

EXPERIENCE

Full-Stack Software Developer — Sensofusion — September 2025 – February 2026

- Fixed 30+ production bugs and shipped UI/UX improvements, including a mobile-view redesign, improving reliability and operator usability.
- Rebuilt and validated the Playwright end-to-end test suite, improving CI reliability and regression coverage.
- Deployed production workloads on AWS (ECS, EKS, Lambda, S3) and configured Cloudflare for secure routing and edge delivery.
- Developed and maintained the command-and-control Next.js application with Redux, integrating multiple sensor types and countermeasure systems.
- Built live telemetry dashboards and geospatial map views (D3.js, Leaflet, Mapbox, WebGL) for operator situational awareness.
- Built Node.js backend-for-frontend (BFF) services to ingest, normalize, and stream sensor and mission data to the C2 client.
- Reduced streamed payload volume by 70% for duplicated notifications through structural backend optimizations.
- Supported Canadian market expansion through on-site field deployments, live customer demonstrations, and hardware/software integration.

Founder / Director — Şandin Tech Inc. — January 2023 – Present

- Lead Şandin Tech end to end—problem validation, product engineering, go-to-market, and startup operations for the Şand peer-to-peer delivery platform.
- Architect and maintain a Next.js/Express/MongoDB platform with modular REST APIs, containerized environments, and CI/CD-ready testing infrastructure.
- Build core product capabilities including JWT authentication, geolocation filtering, user verification, trust-based ratings, and video upload/streaming workflows.
- Built geospatial map views and server-side map tile generation using D3.js, Leaflet, Mapbox, and WebGL to support delivery discovery and workflow UI.

Cloud RAN Software Developer (Co-op) — Ericsson — May 2021 – April 2023

- Developed a cloud-enabled messaging system for radio equipment running on embedded Linux.
- Coordinated integration of software modules with existing radio control systems and legacy interfaces.
- Configured embedded Linux startup and system initialization for radio equipment; optimized software to run on a single processor core.
- Created Jenkins CI/CD pipelines for automated module testing and wrote scripts to automate submodule testing post-merge.
- Migrated legacy unit testing code from C to C++ and automated unit testing through designed test configurations.

PROJECTS

Raspberry Pi Home Assistant Hub

Deployed and maintain a Raspberry Pi-based Home Assistant instance—OS imaging, networking setup, sensor and device integrations, automations, and ongoing administration.
Raspberry Pi OS · Home Assistant · Python · TCP/IP

Autonomous Driving (Queen's AutoDrive)
Worked on 3D point cloud segmentation and classification using ROS for system integration on an autonomous vehicle platform.
Python · ROS · Keras · CNN · Transformers

Facial Emotion & Gesture Recognition
Trained a CNN to classify seven emotions from images (90% accuracy) and designed a live-video gesture recognition system above 20 fps that synthesizes speech from a custom dictionary (70% accuracy).
Python · Flask · OpenCV · Keras · TensorFlow · NumPy

Deepfake Detection
Built and trained a deep neural network to detect deepfake images.
Python · Flask · TensorFlow · Pandas

Elopement Prevention System
Designed a sensor-integrated system to secure nursing home premises using embedded hardware and wireless connectivity.
C++ · Hardware · Sensors · Bluetooth Mesh · RFID · IR

16-bit Microprocessor Design
Simulated a custom processor with branching functionalities and UART integration.
Assembly · VHDL · ModelSim · Altera

SKILLS

Languages C++, Python, TypeScript, JavaScript, C, Java, SQL, bash, Go (familiar), C# (familiar)

Web & APIs Next.js, React, Redux, Node.js, Express, REST, MongoDB, Playwright

Cloud & DevOps AWS (ECS, EKS, Lambda, S3), Docker, Kubernetes, Jenkins, CI/CD, Git

Systems & embedded Embedded Linux, Microcontrollers, Raspberry Pi, TCP/IP, Home Assistant

AI/ML TensorFlow, PyTorch, OpenCV, Computer vision, Scikit-learn

EDUCATION

Bachelor of Applied Science, Computer Engineering (Innovation Stream) — Queen's University
September 2017 – April 2023

Coursework: Operating Systems, Algorithms, Cryptography & Network Security, Microprocessor Systems, Image Processing, Neural & Genetic Computing, Digital Systems Engineering, Software Quality Assurance
Award: First place, mechatronics/robotics competition—built and programmed a robot with fine-tuned servo motion control and multi-sensor feedback (C++, Arduino).

PUBLICATION

Stroke Prediction
December 2021
David Publishing Company
Developed a machine learning model using a Multilayer Perceptron to predict stroke susceptibility.

VOLUNTEER

Chatbot Development (VisaPlace) — QMIND AI Consulting September 2020 – April 2021

Developed an AI chatbot using BotPress to automate applicant processing and scoring for an immigration law firm.

TypeScript · Docker · BotPress · Git